Week No. 1

Lab Manual: Information Security

Lab Title: Classical Ciphers (Code)

Lab Objectives

By the end of this lab, students will be able to:

1. Implement and analyze classical encryption techniques through coding practice.
2. Understand the working principles of the substitution-based ciphers.
3. Evaluate strengths and weaknesses of each classical cipher in modern context.

Learning Outcomes (CLOs & PLOs Mapping)

CLO	CLO Statement	Bloom's Level	PLO
1	Apply Information Security techniques into practice.	P3 (Apply)	PLO-5

Lab Overview

This lab focuses on **Classical Cryptographic Algorithms** — the foundation of modern cryptosystems. Students will study and implement:

- **Caesar Cipher**
- **Vigenère Cipher**
- **Auto-Key Cipher**
- **Playfair Cipher**

Experiment 1: Caesar Cipher

Theory:
A substitution cipher where each letter in the plaintext is shifted by a fixed number (key) within the alphabet.

Encryption Formula:

$$C = (P + K) \bmod 26$$

Decryption Formula:

$$P = (C - K) \bmod 26$$

Pseudo Code:

```
INPUT: Plaintext, Key
FOR each character in Plaintext:
   IF character is alphabet:
      Shift character by Key positions
      Wrap around using mod 26
   ELSE:
      Keep character unchanged
OUTPUT: Ciphertext
```

Task:

- Implement encryption and decryption functions.
- Test with key = 3 on plaintext student's full name.
- Values must not be hardcoded.

Experiment 2: Vigenère Cipher

Theory:
A polyalphabetic cipher using a **keyword** to vary the shift amount per letter.

Encryption Formula:

$$C_i = (P_i + K_i) \bmod 26$$

Decryption Formula:

$$P_i = (C_i - K_i) \bmod 26$$

Pseudo Code:

```
INPUT: Plaintext, Keyword
EXTEND keyword to match length of plaintext
FOR each character in Plaintext:
   Convert both plaintext and keyword letters to numbers (A=0 to Z=25)
   Cipher = (Plain + Key) mod 26
OUTPUT: Ciphertext
```

Task:

- Implement encryption and decryption functions.
- Test with keyword “KEY” on plaintext “CRYPTOGRAPHY”.
- Values must not be hardcoded.

Experiment 3: Auto-Key Cipher

Theory:
A modification of Vigenère Cipher where the **plaintext itself** becomes part of the key after the initial keyword.

Pseudo Code:

```
INPUT: Plaintext, Initial_Key
GENERATE extended_key = Initial_Key + first (len(Plaintext) - len(Initial_Key)) letters of Plaintext
FOR each character:
   Cipher = (Plain + Key) mod 26
OUTPUT: Ciphertext
```

Task:

- Implement encryption and decryption functions.
- Test with keyword “KEY” on plaintext “CRYPTOGRAPHY”.
- Values must not be hardcoded.

Experiment 4: Playfair Cipher

Theory:
A **digraph substitution cipher** using a 5x5 matrix constructed from a keyword.

Steps:

1. Create 5x5 matrix from keyword (merge I/J).
2. Split plaintext into digraphs.
3. Apply following rules:
    - Same row → replace with right letter.

- Same column → replace with below letter.
- Otherwise → replace with letters at rectangle corners.

Pseudo Code:

```
INPUT: Keyword, Plaintext
GENERATE 5x5 matrix removing duplicates, merging I/J
SPLIT plaintext into pairs (add X if needed)
FOR each pair:
   IF same row: replace each with next right letter
   ELSE IF same column: replace each with next below letter
   ELSE: replace each with letter in same row, other column
OUTPUT: Ciphertext
```

Task:

- Implement encryption and decryption functions.
- Test with keyword “MONARCHY” on plaintext “CRYPTOGRAPHY”.
- Values must not be hardcoded.

Class Participation Task (20 Marks)

You are required to design a hybrid encryption process using two sequential approaches:

Approach 1: Layered Classical Encryption

Take any key word (e.g., “SECURE”).

Encrypt this key using Caesar Cipher with a shift of 3.

Example: “SECURE” → Caesar(+3) → “VHF XUH”

Use the resulting Caesar-encrypted key as the Playfair cipher key.

Encrypt your chosen plaintext (e.g., your name) using the Playfair Cipher matrix generated from this new key.

Observe how the first cipher (Caesar) influences the behavior of the second.

Approach 2: ASCII-Based Hybrid Cipher

Convert all characters (both plaintext and key) to their ASCII values.

Perform Caesar-like shifting using numerical operations on ASCII codes:

$$C = (P + K) \text{mod } 128$$

Now use the resulting ASCII sequence as input for the Playfair Cipher, but replace alphabetic mapping with ASCII range (0–127).

Discuss:

How does using ASCII instead of alphabets affect cipher strength?

What new challenges arise in decryption?